Practice Problems Unit 5
AP Calculus Name_____

The following problems are sample problems from recent units.

Given the table of values, find the following:

x	$g(x)$	$g'(x)$	$h(x)$	$h'(x)$
3	-1	7	-2	-3
4	3	-2	9	5

1.) $f'(4)$ for $f(x) = (g(x))^2$

2.) $p'(4)$ for $p(x) = \sqrt{h(x)}$

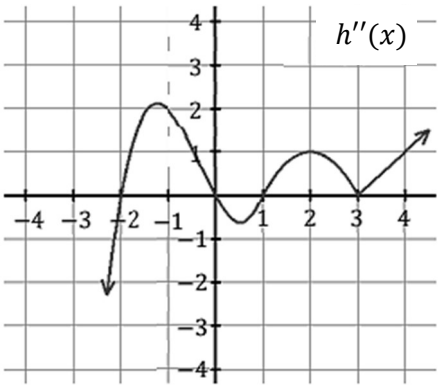
3.) (a) $r'(4)$ for $r(x) = h(g(x))$

(b) the equation of the tangent line to $r(x)$ at $x = 4$

4.) The graph of $h''(x)$ is at the right. No justification is needed for this problem.

(a) Find all x values of the points of inflection of $h(x)$.

(b) On what intervals is $h(x)$ concave up? Concave down?

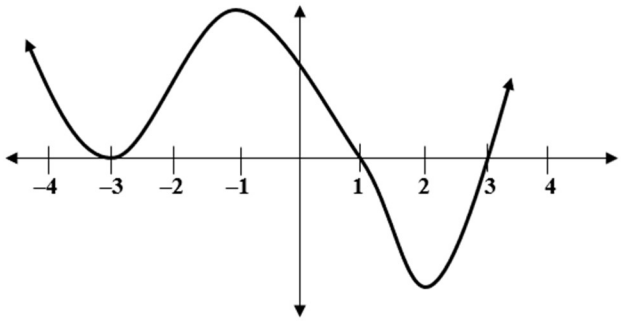


5.) Given $h'(x) = 2x^2(x+5)(x-7)$, find the following:

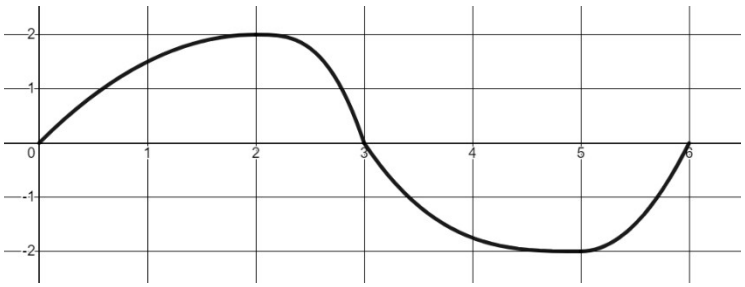
(a) the critical values of $h(x)$

(b) the x-values of any relative extrema

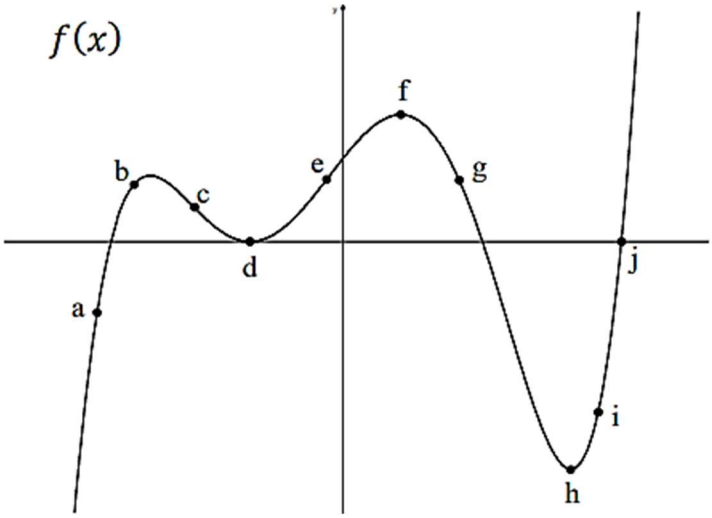
6.) The graph below is the graph of the derivative, f ′(x), of a function whose domain is (–∞, ∞) and has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 2$. For what values of x is the function f(x) increasing? Decreasing? Concave up? Concave down? x-values of the points of inflection? JUSTIFY YOUR ANSWERS!



7.) The graph of the velocity, v(t), of a particle on [0, 6] is given below. The velocity function has horizontal tangents at $x = 2$ and $x = 5$. When is the particle moving to the right? to the left? When is the acceleration positive? negative? When is the particle speeding up? slowing down? JUSTIFY YOUR ANSWERS (USING THE VELOCITY FUNCTION)!



8.) Using the figure at the right, complete the chart by indicating whether each value is positive (+), negative (–), or zero (0) at the indicated points. For these problems, if the point appears to be a max or min, assume it is. If it appears to be a point of inflection, assume it is.



x	A	B	C	D	E	F	G	H	I	J
$f(x)$										
$f'(x)$										
$f''(x)$										